

5. TRANSPORTATION

MBS

NUMERICAL QUESTIONS

1. 2071 Q.No. 8

Four factories A, B, C and D produces sugar and the demand of each warehouse X, Y, Z are given below: [10]

Factories	Cost in Rs. per ton ('000)			Availability in tons
	X	Y	Z	
A	4	3	2	10
B	5	6	1	8
C	6	4	3	5
D	3	5	4	6
Requirement in tons	7	12	4	23

Find the optimal solution for the least transportation cost.

Ans: Total minimum transportation cost = Rs. 65,000

2. 2070 Q.No. 9

A national truck rental firm is planning for a heavy demand of rental trucks during the month of June. From the following data on number of trucks to be allocated and the cost involved, find optimum solution to minimize the transportation cost. [10]

Origin (surplus area)	Destination (shortage area)			Supply of trucks
	1	2	3	
1	50	100	100	110
2	200	300	200	160
3	100	200	300	150
Demand of trucks	140	200	80	420

Ans: Rs. 67,000

3. 2070 Old Q.No. 8a

Determine optimum transportation model and find minimum cost of transportation. [10]

Stores	Ware house			Requirements
	W ₁	W ₂	W ₃	
	(Cost per unit)			
S ₁	45	35	30	60
S ₂	60	15	25	40
S ₃	45	35	45	60
S ₄	30	35	55	20
Availability	70	60	90	

Ans: Rs. 5,500

4. 2069 Q. No. 9

Solve the transportation problem. [10]

Destination Origin	D ₁	D ₂	D ₃	Available
O ₁	30	20	10	800
O ₂	5	15	25	500
Required	300	300	400	

Ans: Total minimum transportation cost = Rs. 10,500

5. 2069 (Old) Q. No. 8a

The following table represents the profit per unit of goods. Find optimal transportation schedule so that the profit is maximized.

Demand centres	Supply centres			Requirements
	S ₁	S ₂	S ₃	
D ₁	1300	1700	1800	3000
D ₂	1100	1400	1800	3000
D ₃	1500	1200	1500	4000
D ₄	2000	1300	1200	5000
Availability	2000	6000	7000	15000

Ans: Total maximum profit = Rs. 2,44,00,000

6. 2068 Old Q.No. 8a

Find the optimum transportation schedule to minimize the cost from the following [10]

Plant	Stores				Available quantity
	S ₁	S ₂	S ₃	S ₄	
P ₁	90	120	90	60	70
P ₂	70	30	70	70	60
P ₃	60	50	90	110	90
Required quantity	60	40	60	20	220 180

Ans: Total minimum transportation cost = Rs. 11,000

7. 2067 Q.No. 9

Find optimal transportation schedule to minimize the cost from the following unbalanced transportation problem. [10]

Plant	Stores				Available quantity
	S ₁	S ₂	S ₃	S ₄	
W	29	32	29	26	70
X	27	23	27	27	60
Y	26	25	29	31	90
Demand units	60	40	60	20	

Ans: Total min transport cost = Rs. 4,700

8. 2067 Q.No. 8a (Old)

Using transportation problem, determine the maximum profit from the information given below.

Ware house	Factory			Capacity
	Profit in Rupees			
	X	Y	Z	
A	130	170	180	30
B	110	140	180	30
C	150	120	150	40
D	200	130	120	50
Demand	20	60	70	

Ans: Max Total profit = Rs. 24,400

9. 2066 Q.No. 7 b

Determine the optimum solution for the following problem to minimize the transportation cost: 10

	W ₁	W ₂	W ₃	Supply
P ₁	3	8	5	5
P ₂	5	5	3	8
P ₃	7	8	9	7
P ₄	4	9	5	14
Requirements	7	9	18	34

Ans: Minimum transportation cost = Rs. 167 and $X_{11} = 5$ units; $X_{22} = 2$ units; $X_{23} = 6$ units; $X_{32} = 7$ units; $X_{41} = 2$ units; $X_{43} = 12$ units.

10. 2065 Q.No. 9 a

Find the optimum solution that minimized the transportation cost for following data: [10]

	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	8	8	10	9	15
O ₂	8	12	10	10	25
O ₃	12	10	11	11	15
Requirements	15	15	15	10	55

Ans: $X_{12} = 15$; $X_{21} = 15$; $X_{23} = 10$; $X_{32} = 5$; $X_{34} = 10$; Min. cost = Rs. 505

11. 2064 Q.No. 8 a

Find the optimum solution for the problem below that minimizes the total transportation cost. [10]

	D ₁	D ₂	D ₃	D ₄	Supply
P ₁	2	3	11	7	6
P ₂	1	6	6	1	1
P ₃	5	8	15	9	10
Requirements	7	5	3	2	17

Ans: Rs. 100

12. 2062 Q.No. 9 (b)

'Nepal Transport' own several trucks used to deliver crushed stones to road projects in the region. The company has received the delivery schedule for next week as follows:

Project	Requirement per week	Plant	Available per week
A	50	W	55
B	75	X	60
C	50	Y	60

The cost matrix for above schedule is given as follows:

Plant	Cost information ('000 Rs.) Project		
	A	B	C
W	4	8	3
X	6	7	9
Y	8	2	5

Find an optimum solution to minimize the transportation cost.

[10]

Ans: Minimum transportation cost = Rs. 6,65,000

13. 2061 Q.No. 8 (a)

Determine the minimum transportation cost from the following matrix.

[10]

Warehouse	Stores				Supply
	P1	P2	P3	P4	
	Cost per unit				
W1	45	60	45	30	70
W2	35	15	35	35	60
W3	30	25	45	55	90
Demand	60	40	60	20	220
					180

Ans: Minimum transportation cost = Rs. 5,500

14. 2060 Q.No. 8 (a)

From the following profit matrix, find the maximum profit by using transportation model.

Plant	Units Available	Project	Units demanded
A	1700	W	1300
B	2500	X	2000
C	1000	Y	1900

Profit in '000 rupees

[10]

From	To Project W	To Project X	To Project Y
Plant A	12	8	5
Plant B	11	15	10
Plant C	2	17	6

Ans: Maximum profit = Rs. 64,600

15. 2059 Q.No. 9 (b)

'Nepal Hardware Company' has three plants in Kathmandu, Banepa and Nuwakot and three warehouses at Bhairahawa, Birgunj and Biratnagar. The quantity available at the plants are respectively 60, 70 and 80, where as the demand at the warehouses are 50, 80 and 80 respectively. The unit cost of transportation are observed as follows:

	Bhairahawa	Birgunj	Biratnagar
Kathmandu	2	5	7
Banepa	2	3	4
Nuwakot	5	8	11

Find an optimum allocation that minimizes the transportation cost.

[10]

Ans: Minimum transportation cost = Rs. 1,090

16. 2058 Q.No. 8 (a)

Find the optimal transportation schedule from following with the objective of minimizing the cost.

[10]

Factory	Quantity requirements per day in kg.	Ware house	Quantity available per day in kg.
P	450	X	350
Q	500	Y	400
R	200	Z	400

Cost of transportation per kg is given in the following table.

From	To factory		
	P	Q	R
Warehouse X	10	20	20
Warehouse Y	40	60	40
Warehouse Z	10	16	24

Ans: Minimum transportation cost Rs. 26,900

MBA

17. 2055 Q.No. 7 (a)

Find optimal transportation schedule to minimize the cost from the following information. [10]

Plants	Stores					Capacities
	S ₁	S ₂	S ₃	S ₄	S ₅	
P ₁	7	10	5	4	12	45
P ₂	3	2	0	9	1	70
P ₃	8	13	11	6	14	125
Order size	10	20	30	80	100	240

Ans: Minimum transportation cost = Rs. 1,415

18. 2053 Q.No. 7 (a)

Suggest an optimal transportation plan with a view to minimize the cost and determine the minimum transportation cost from the following: [10]

Warehouse	Stores (Cost per units in Rs.)				Supply
	S ₁	S ₂	S ₃	S ₄	
W ₁	9	12	9	6	7
W ₂	7	3	7	7	6
W ₃	6	5	9	11	9
					22
Demand	6	4	6	2	18

Ans: Minimum transportation cost = Rs. 110

19. 2052 Q.No. 6 (a)

The Rent-a-car Company rents car trailers to individuals making one way moves. Occasionally the company has to redistribute the trailers in order to eliminate a surplus build up in some cities and a shortage in others. The company currently has 4 trailers in A, 3 in B, 6 in C and 1 in D. They would like to shift 5 trailers to E, 3 to F and 6 to G. The following table gives the mileages between the various sources and destinations.

Sources	Destination		
	E	F	G
A	60	50	80
B	30	40	90
C	60	100	100
D	90	70	30

How should company redistribute the trailers so that total mileage travelled is minimized? [10]

Ans: Minimum mileage = 750

20. 2050 Q.No. 6 (a)

T.C. Mellott trucking company has a contract to move 115 truck loads of sand per week between three sand-washing plants W, X and Y and three destinations, A, B and C. Cost and volume information is given below. [10]

Compute the optimal transportation cost

Project	Requirement per week truck loads	Plant	Available per week truck loads
A	45	W	35
B	50	X	40
C	20	Y	40

Cost information:

From	To project A	To project B	To project C
Plant W	Rs. 5	Rs. 10	Rs. 10
Plant X	20	30	20
Plant Y	5	8	12

21. 2045 Q.No. 6 (a)

Ans: Minimum transportation cost = Rs.1,345

A company has three factories from where 2000 cases have to be transported to five warehouses. The demands of the warehouses, the supplies available at the different factories, and the transportation costs per case are given in the following matrix.

Warehouse	Factory			Demand
	1	2	3	
1	0.50	0.65	0.54	320
2	0.48	0.55	0.60	470
3	0.72	0.60	0.58	440
4	0.53	0.51	0.57	350
5	0.45	0.49	0.59	420
Supply	900	500	600	2000

Find the least cost distribution schedule.

[10]

Ans: Minimum transportation cost = Rs. 1,020.70

22. 2042 Q.No. 6 (a)

Suggest an optimal transportation plan with a view to minimize costs from the following information: [10]

Destination	Sources			Units Demanded
	F ₁	F ₂	F ₃	
W ₁	Rs. 0.90	Rs. 1.00	Rs. 1.30	5
W ₂	Rs. 1.00	Rs. 1.40	Rs. 0.80	20
W ₃	Rs. 1.30	Rs. 1.00	Rs. 0.80	20
Units available	20	15	10	45

Ans: Minimum transportation cost = Rs. 42.5

23. 2040 Q.No. 9 (a)

Find the optimal transportation schedule on the basis of the following information:

[10]

Factory	Stores (Haulage cost in Rs.)				Availability
	S ₁	S ₂	S ₃	S ₄	
F ₁	9	7	10	8	14
F ₂	8	11	9	11	27
F ₃	13	10	12	10	14
Requirement	15	19	11	10	55

Ans: Minimum transportation cost = Rs. 468

24. 2039 Q.No. 9 (a)

ABC Company supplies raw materials to three different factories X, Y and Z from its three different warehouses A, B and C. Cost and volume information are given below. Compute the optimal transportation cost.

[10]

Factory	Requirement per week truck loads	Warehouse	Available per week truck loads
X	45	A	35
Y	50	B	40
Z	20	C	40

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Cost information:

From	To factory X	To factory Y	To factory Z
Warehouse A	Rs. 50	Rs. 100	Rs. 100
Warehouse B	200	300	200
Warehouse C	50	80	120

Ans: Minimum transportation cost = Rs. 13,450